11/06 From CSV to PCAP

CSV inputs showed clear limits, so features are now built directly from raw packets

Prior experiments	Problems found	This report
CICFlowMeter CSV vectors	Weak classification	PCAP instead of CSV
XGB · TabPFN SOTA · proposed models	Same features, different labels	Features from netFound, a transformer encoder for network data
Evaluation scenarios attempted	Some attacks score high even in few-shot, leaving little to gain	

Question Do packet-based features fix these problems?

Problem 1 Same Features, Different Labels

PCAP is captured in a testbed, then CICFlowMeter groups packets into flows and computes fixed statistics

Testbed	PCAP	CICFlowMeter	CSV
Benign and attack traffic	Raw packets in time order	Group flows · fixed statistics	Statistics vector + label

Fixed statistics the same 78 formulas per flow · duration · packets sent, received · mean packet length · mean gap · FIN count

How the same features get different labels

① Main cause: flow boundary	② Very short flows	③ Labels follow the parent
Cuts at the first FIN, ignores RST Leftover ACK and FIN become flows These make up 25.9% of CIC2017	Statistics from 1 or 2 packets easily match	Identical tail flows carry benign and attack labels

Evidence Engelen 2021 · Liu 2022 · In our PCAP rebuild, removing tail flows cuts statistics conflicts from 6,367 to 69

Data	Flows with identical statistics	Labels attached
CIC2017	Dst port 80 · 1 µs · sent 2 · received 0	BENIGN 2 rows · DoS Hulk 3,360 rows
CIC2018	Dst port 21 · 2 µs · sent 1 · received 1	SlowHTTPTest 62,024 rows · FTP 87,297 rows · SSH 10 rows

Problem 1 Conflicts by Class in the CSVs

All rows of the original CSVs · Rows in groups with identical statistics but different classes · Ordered by CIC2017 rows

class	CIC2017 rows	Conflicts	Share %	CIC2018 rows	Conflicts	Share %
benign	2,273,097	736	0.03	13,484,708	125,771	0.93
dos	252,661	5,225	2.07	654,300	77,277	11.81
portscan	158,930	1,053	0.66	–	–	–
ddos	128,027	6	0.00	1,263,933	0	0.00
bf	13,835	0	0.00	380,949	162,479	42.65
web	2,180	0	0.00	928	5	0.54
bot	1,966	0	0.00	286,191	0	0.00
inf	36	0	0.00	161,934	41,279	25.49
heartbleed	11	0	0.00	–	–	–
Total	**2,830,743**	**7,020**	**0.25**	**16,232,943**	**406,811**	**2.51**

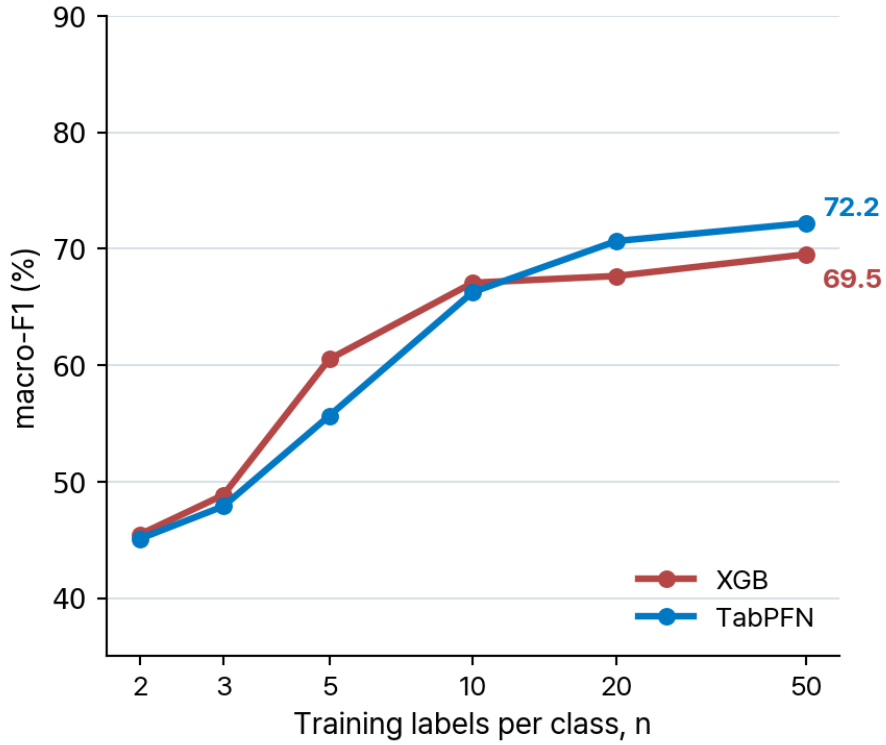
CIC2018 attack classes conflict heavily: bf 42.65%, inf 25.49%, dos 11.81% · CIC2017 peaks at dos 2.07%

Experiments use CIC2017 because of PCAP size (CIC2018 450 GB, CIC2017 51 GB)

Problem 2 Some Attacks Score High Even in Few-Shot

CIC2017 CSV · Attacks only in training and evaluation · n labels per class · Mean of 5 seeds

Macro-F1 over six attacks



F1 by class (%) · TabPFN

	n=2	n=3	n=5	n=10	n=20	n=50
dos	59	66	83	88	92	95
portscan	77	70	70	98	98	98
ddos	82	87	87	89	94	95
bf	22	29	58	79	84	90
bot	0	0	0	0	0	0
web	30	36	36	44	56	55

At n=20: portscan 98%, ddos 94%, dos 92% · At n=50: 98%, 95%, 95%

web stays at 55% at n=50, and bot 0% is a CSV label defect · Mean scores alone cannot show difficulty

From PCAP to Features

Labels are not used to compute features · The time split comes first so PCA is fit on training data only

① PCAP

CIC2017 Mon to Fri
One file per day
5 files, 51 GB

② Group into flows

Same protocol, IPs and ports
Ends at RST · ACK after both FINs
new SYN · 120 s

③ Label

Attacker and victim IPs?
Inside the attack window?
Evidence of attack behavior?

④ Time split

By day and attack
First 70% train
Next 15% val · last 15% test

⑤ Tokenize

IP hdr len, TOS, length, flags, TTL
TCP flags, window, seq, ack, urgent
No IPs, ports, times or payload

⑥ netFound

Tokens → 512 dims
Frozen weights

⑦ PCA

512 → 64 dims
Fit on training data

⑧ Evaluate

Packets become labelled flows, then header tokens pass through the encoder to give 64 features per flow

What a Packet Looks Like

A packet is one piece of data on the network · Below: a real 151-byte packet from an office PC to a web server



Ethernet 14 B	IP header 20 B	TCP header 20 B	payload 97 B
Delivery to the next device Source device Destination device	Which computer Source 192.168.10.9 Destination 69.31.33.224 Total length 137 B TTL 128 Protocol 6 (TCP)	Which program Source port 1057 Destination port 80 seq 2039531092 ack 2775650109 Flags ACK+PSH window 256	The actual content GET /ncsi.txt HTTP/1.1 Connection: Close User-Agent: Microsoft NCSI Host: www.msftncsi.com

A packet has headers outside and payload inside, and this work does not read payload content

Grouping One Conversation into a Flow

A flow is the packets two programs on two computers exchange under one protocol: one conversation

Protocol	IP A	Port A	IP B	Port B

If all five values match

Same flow · even when sender and receiver swap

Packet	Protocol	From	To	Result
Request	**TCP**	**192.168.10.9 : 1057**	**69.31.33.224 : 80**	**flow A**
Response	**TCP**	**69.31.33.224 : 80**	**192.168.10.9 : 1057**	**flow A direction reversed**
Auth 1	TCP	192.168.10.9 : 1055	192.168.10.3 : 88	flow B
Auth 2	TCP	192.168.10.9 : 1056	192.168.10.3 : 88	flow C different temporary port
DNS lookup	UDP	192.168.10.9 : 59285	192.168.10.3 : 53	flow D other protocol and peer

Why group One packet says little; the whole conversation shows the request, response and end

Once a conversation ends, the next packet starts a new flow · End conditions follow

Where a Flow Ends: FIN and RST

→ PC sends ← server sends · The last column shows how each packet is handled · Real Monday packets

Case 1 Ends with FIN · 192.168.10.9:1057 ↔ 69.31.33.224:80

#	Time (µs)	Dir	TCP flag	Len	payload	Flow
1	0	→	SYN	52	0	flow A
2	19,958	←	SYN+ACK	52	0	flow A
3	20,008	→	ACK	40	0	flow A
4	20,083	→	ACK+PSH	137	97	flow A
5	40,027	←	ACK	40	0	flow A
6	40,640	←	ACK+PSH	219	179	flow A
7	40,643	←	ACK+FIN	40	0	flow A
8	40,685	→	ACK	40	0	flow A
9	40,689	→	ACK+FIN	40	0	flow A
10	**60,574**	**←**	**ACK**	**40**	**0**	**flow A closes**

Case 2 Ends with RST · 192.168.10.51:38488 ↔ 209.15.36.31:80

#	Time (µs)	Dir	TCP flag	Len	payload	Flow
1	0	→	SYN	60	0	flow B
2	42,246	←	SYN+ACK	60	0	flow B
3	42,350	→	ACK	52	0	flow B
4	5,079,690	→	ACK+FIN	52	0	flow B
5	5,084,903	←	ACK+FIN	264	212	flow B
6	**5,084,965**	**→**	**RST**	**40**	**0**	**flow B closes**
7	**5,121,931**	**←**	**ACK**	**52**	**0**	**dropped**
8	**5,121,962**	**→**	**RST**	**40**	**0**	**dropped**

The PC closed first with FIN at 4, so it answers 5 and 7 with RST

The flow ends at the first RST · Later tail packets are dropped

TCP flags

SYN open · ACK acknowledge
PSH deliver now · FIN done sending
RST abort at once, no reply needed

Dropped tail packets

Arrive within 2 s of the close
At most 2 packets, no SYN or payload

When a flow ends

RST · ACK after both FINs · New SYN
120 s idle · 120 s since start
UDP and ICMP use time only

A FIN flow runs to the last ACK, an RST flow to the first RST, and later packets are dropped

From Header Tokens to netFound Features

Each packet header becomes numeric tokens fed to netFound · Below: packets of flow A in case 1

Packet	IP hdr len	TOS	Total len	IP flag	TTL	TCP flag	window	seq high	seq low	ack high	ack low	urgent
Packet 1 SYN	20	0	52	2	128	2	8192	0	0	0	0	0
Pkt 2 SYN+ACK	20	0	52	2	56	18	29200	0	0	0	1	0
Packet 4 request	20	0	137	2	128	24	256	0	1	0	1	0
…	…	…	…	…	…	…	…	…	…	…	…	…

Header tokens of one flow · 1 packet = 12 numbers · No IPs, ports, times or payload content

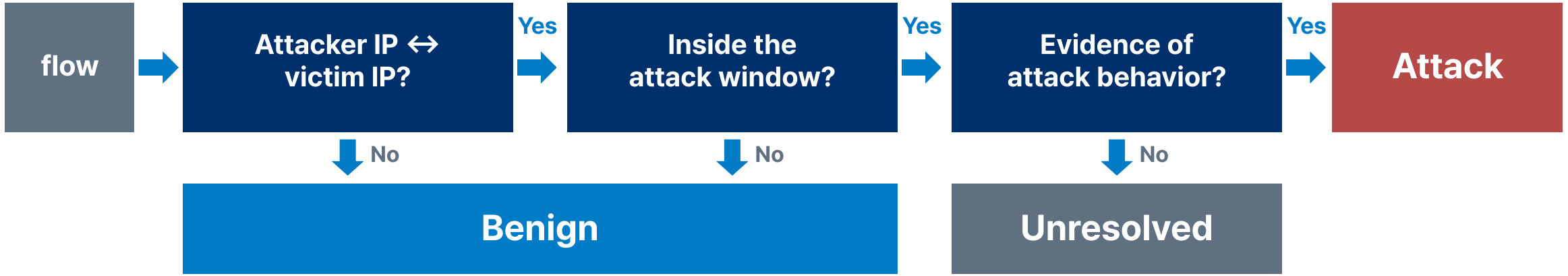
Header tokens	netFound	PCA	feature
In packet order, with gaps and direction	Pretrained transformer Frozen weights	512 → 64 dims Fit on training data	One flow = 64 values One classifier row

netFound a transformer pretrained on network traffic

Learns by predicting masked header tokens in unlabelled traffic, used here as an encoder only

Labelling Flows

Each flow is judged from left to right · The 53 corrected rules of CNS 2022 are applied as published



Label	Condition	Handling
Attack	IPs, window and evidence all met	Used as attack
Unresolved	IPs and window match, no evidence	Excluded from train and test
Benign	IPs or window outside the rules	Used as benign

Evidence FTP and SSH brute force: attacker payload ≥ 1 B

Evidence Portscan: one SYN with no reply or RST

CNS 2022 Liu et al.

Error Prevalence in NIDS Datasets

Packet-level study of flow and label errors in CIC-IDS 2017 and 2018
Published corrected rules and tools

Final Data

After deleting residual flows, before the split · CSV: official label rows; PCAP: labelled flows · Ordered by PCAP flows

class	Attacks (PCAP flows)	CSV rows	PCAP flow
benign	All of Monday + background of attack days	2,273,097	1,609,980
portscan	Fri Portscan 161,202 · Thu NMAP 76,947	158,930	238,149
dos	Hulk 158,546 · GoldenEye 7,567 · Slowloris 3,859 · Slowhttptest 1,741	252,661	171,713
ddos	LOIC-HTTP	128,027	95,683
bf	FTP 4,021 · SSH 2,961	13,835	6,982
bot	Ares	1,966	736
web	Brute Force 73 · XSS 18 · SQL Injection 13	2,180	104
inf	Communication 32 · CoolDisk 4	36	36
heartbleed	Heartbleed	11	11
Total		**2,830,743**	**2,123,394**

Counts differ from CSV because flow boundaries and labels are corrected · Thursday NMAP is BENIGN in CSV, portscan here

inf has 25 train and 6 test flows, and heartbleed has 7 train and 2 test flows

Problem 1 Revisited: Label Conflicts

After deleting residual flows · Same features, different classes · Concentrated in very short flows

feature	Conflicts	Share
CSV 78 statistics	7,020	0.25%
netFound	**43,380**	**2.04%**

Remaining Portscan 42,609 · benign 770 · DoS 1

Packets	Flows	Same input %	Conflicts
1	7,853	97.2	821
2	**505,950**	**97.2**	**42,443**
3~5	691,512	92.1	110
6~20	608,631	11.1	6
21 or more	309,401	0.8	0

Causes

Very short flows have few numbers, mostly fixed
Example scan SYN and refusing RST, 2 packets

Information netFound drops before input
IPs · ports · times · payload content

Side information can help

Columns behind the netFound vector, without IP or port values

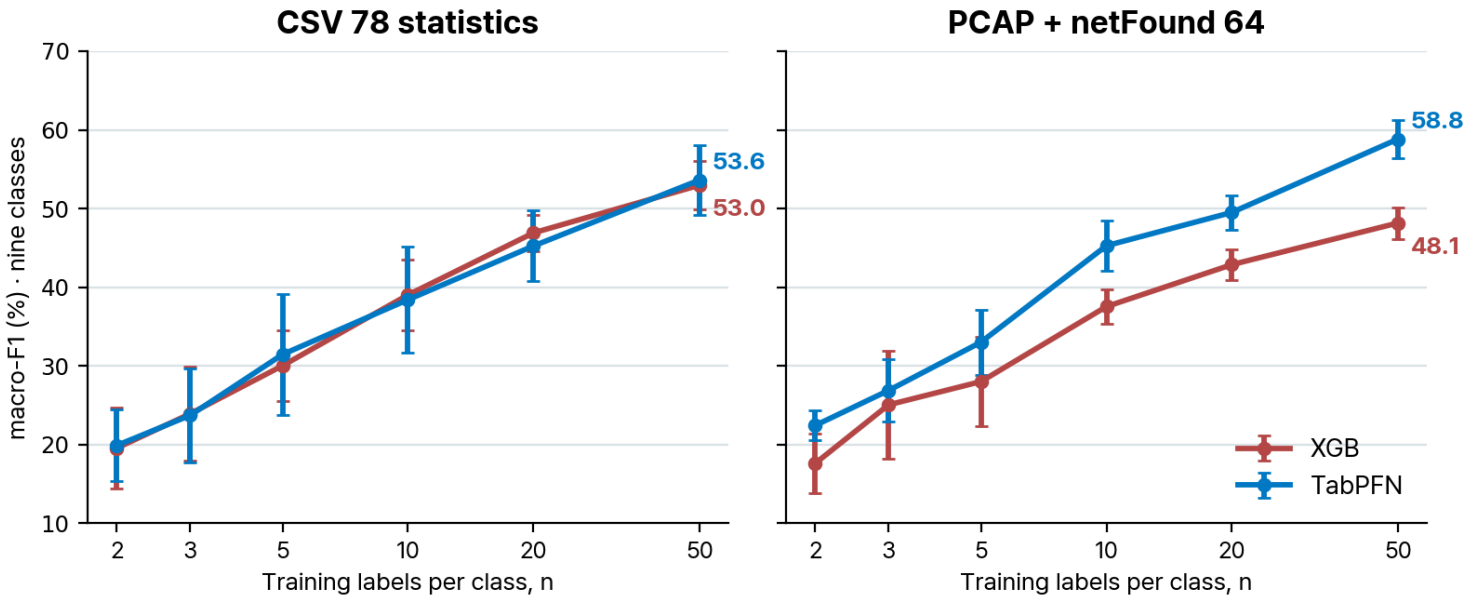
Input	Conflicts	Share	Reduced
netFound	43,380	2.04%	–
+ raw seq flag	42,571	2.00%	1.9%
+ ports in the last 60 s	964	0.05%	97.8%
+ both	**302**	**0.01%**	**99.3%**

The 60 s port count alone cuts conflict flows by 97.8%

The remaining 2.04% comes from very short flows and dropped fields; side information cuts it to 0.05%

Problem 2 Revisited: Few-Shot Performance

All nine classes · Time split · 5 seeds · Same training rows · netFound after deleting residual flows



macro-F1	XGB n=20	TabPFN n=20	XGB n=50	TabPFN n=50
CSV	46.9	45.2	53.0	53.6
netFound	42.9	49.5	48.1	58.8

Difference from CSV

Both models are close on CSV
TabPFN leads clearly on netFound
Deletion: −0.5 pp (7 classes, n=50)

Why TabPFN leads n=50, before deletion

Rows equal to training inputs: both models 97.0%
Other rows: XGB 91.1% · TabPFN 97.5%
The gap comes from unseen inputs
PCA vectors suit axis splits poorly (inferred)

netFound + TabPFN n=50 F1 by class

benign 97.9 · portscan 97.3 · dos 89.9 · ddos 99.1 · bf 94.6
bot 35.2 · web 10.0 · inf 0.2 · heartbleed 5.1
inf and heartbleed: 6 and 2 test flows, high recall, low precision

Few-Shot with Side Information

Nine-class macro-F1 · Columns behind the netFound vector · 5 seeds · Same training rows · Residual flows deleted

Input	XGB n=20	XGB n=50	TabPFN n=20	TabPFN n=50	val gain pp TabPFN n=50	False alarm % TabPFN n=50	Portscan F1 XGB n=50
netFound	42.9	48.1	49.5	58.8	–	2.97	94.2
+ raw seq flag	**43.5**	48.4	50.1	59.2	−0.4	3.18	94.9
+ ports in the last 60 s	43.1	**48.8**	**50.4**	**60.2**	**+2.3**	**2.48**	**98.5**

Ports in the last 60 s

TabPFN n=50 +1.4 pp, up in 4 of 5 seeds
Validation also +2.3 pp

Benign false alarms 2.97 → 2.48%
XGB Portscan F1 94.2 → 98.5

Raw seq flag

TabPFN n=50 +0.4 pp, up in 2 of 5 seeds
Validation −0.4 pp

Changes stay within seed spread
so the effect is unclear

Ports in the last 60 s help slightly on validation and test; the raw seq flag shows no clear effect

Summary and Next Steps

Problem 1 Label conflicts

Deleting residual flows
4.86 → 2.04%

Remaining: short scan flows
60 s port count: down to 0.05%

Problem 2 Few-shot

Few-shot still works well
Nine-class n=50: 58.8%, CSV 53.6%

High-volume attacks saturate early
bot, web and inf remain low

Why PCAP directly

Correct flow boundaries and labels

Use packet order and context

Experiment with input design

Next

Data	Find more datasets available as PCAP
Models	Test SOTA and proposed models on netFound features
feature	Compare models with the 60 s port count in the input